FUN-6.C.1 $\int f(x)dx$ is an indefinite integral of the function f and can be expressed as $\int f(x)dx = F(x) + C$, where $F'(x) = f(x)$ and C is any constant.
FUN-6.C.2 Differentiation rules provide the foundation for finding antiderivatives.
FUN-6.C.3 Many functions do not have closed-form antiderivatives.
FUN-6.B.3 If f is continuous on the interval $[a, b]$ and F is an antiderivative of f, then $\int_a^b f(x)dx = F(b) - F(a)$

6-8 & 6-7 Antiderivatives, Indefinite Integrals, and Definite Integrals

We spent much of our time early in the year learning and applying derivative rules. Now, we will look at finding **integrals** or **antiderivatives**.

Remember: Derivatives and Antiderivatives are **reverse operations** so these ideas will be connected. Be careful to think first when working problems!

Get Ready...

Given the following functions, find $f'(x)$.

A.) $f(x) = x^2$

$f'(x) = 2x$

B.) $f(x) = \sin(x)$

$f'(x) = \cos x$

C.) $f(x) = 3$

$f'(x) =$

Now, let's think about this in "reverse".
If $g'(x) = 2x$ what do you think would be the equation for $g(x)$?

$$g(x) = x^2 + C$$

Example 1: Given $h'(x)$ below, use your understanding of derivatives to find $h(x)$. You may have to use trial and error on a few of these!

a.) $h'(x) = \cos(x)$

$h(x) = \sin x + C$

b.) $h'(x) = -2$

$h(x) = -2x + C$

c.) $h'(x) = 3x^2$

$h(x) = x^3 + C$

d.) $h'(x) = 0$

$h(x) = C$

e.) $h'(x) = -\sin(x)$

$h(x) = \cos x + C$

f.) $h'(x) = 4x^2$

$h(x) = \frac{4}{3}x^3 + C$

Antidifferentiation (Integration)

Going from $f'(x)$ to $f(x)$ is critical to calculus. We call this process **antidifferentiation** or **integration.**

Antiderivative or Integral Notation
• Instead of writing what function would have a derivative of $f(x)$, we write $\int f(x)\,dx$ which is read " the integral of $f(x)$ with respect to x."
• Generally, we say $\int f(x)dx = F(x) + C$
Note: You MUST ALWAYS write + C after your integral's answer

Indefinite Integral

All of the following formulas are "must know" integral formulas. They are all in terms of "u". This will lead to **u-substitution** later, which is integration's equivalent to **chain rule** with derivatives.

Basic Antiderivative Rules	
$\int k\,du = ku + C$	(k is a constant)
$\int u^n\,du = \frac{1}{n+1}u^{n+1} + C$	(where $n \neq -1$) This is the Power Rule
$\int u^{-1}\,du = \int \frac{1}{u}\,du = \ln\lvert u\rvert + C$	

Two Important Tips for Integration (DO THESE BEFORE USING ANY OF THE RULES ABOVE!!!!)
1. Turn all radicals into powers 2. Move all variables out of the denominator and into the numerator, if possible.

Example 2:

a. $\int 5x^3 dx = \frac{5}{4}x^4 + C$

b. $\int \frac{2}{3}x^2 dx = \frac{1}{3} \cdot \frac{2}{3}x^3 + C$

$= \frac{2}{9}x^3 + C$

c. $\int \frac{4x^5}{3} dx = \int \frac{4}{3} \cdot x^5 dx$

$= \frac{1}{6} \cdot \frac{4}{3}x^6 + C$

$= \frac{2}{9}x^6 + C$

d. $\int 3dx = 3x + C$

e. $\int 1dy = y + C$

Example 3:

a. $\int 5\sqrt{x}dx = \int 5x^{1/2}dx = \frac{2}{3}\cdot 5x^{3/2} + C = \frac{10}{3}x^{3/2} + C$

OR $\frac{10}{3}\sqrt{x^3} + C$

b. $\int \frac{2}{5}\sqrt[4]{x}dx = \int \frac{2}{5}x^{1/4}dx = \frac{4}{5}\cdot\frac{2}{5}x^{5/4} + C$

$= \frac{8}{25}x^{5/4} + C$

$\frac{1}{4}+1$

Example 4:

a. $\int \frac{1}{x^3}\,dx = \int x^{-3}\,dx = \frac{1}{-2}x^{-2} + C = \frac{-1}{2x^2} + C$

b. $\int -\frac{9}{5x^4}\,dx = \int \frac{-9}{5} \cdot x^{-4}\,dx = \frac{-1}{3} \cdot \frac{-9}{5} x^{-3} + C$

$= \frac{3}{5}x^{-3} + C$ OR $\frac{3}{5x^3} + C$

Example 5:

a. $\int \frac{2}{\sqrt{x}}dx = \int 2x^{-1/2}\,dx = 2 \cdot 2x^{1/2} + C$

$= 4x^{1/2} + C$ OR $4\sqrt{x} + C$

b. $\int \frac{-1}{3\sqrt[5]{x}}dx = \int -\frac{1}{3} \cdot x^{-1/5}\,dx = \frac{5}{4} \cdot \frac{-1}{3} x^{4/5} + C$

$= \frac{-5}{12} x^{4/5} + C$

Integration of Trigonometric Functions	
$\int \sin u \ du = -\cos u + C$	$\int \cos u \ du = \sin u + C$
$\int \sec^2 u \ du = \tan u + C$	$\int \csc^2 u \ du = -\cot u + C$
$\int \sec u \tan u \ du = \sec u + C$	$\int \csc u \cot u \ du = -\csc u + C$

Example 6:

a. $\int -\cos y \, dy =$ $-\sin y + c$

b. $\int 3 \sec^2 x \, dx =$ $3 \cdot \tan x + C$

Integration of Exponential Functions	
Base e	$\int e^u \, du = e^u + C$
Any other base (a is a constant)	$\int a^u \, du = \left(\frac{1}{\ln a}\right) a^u + C = \frac{a^u}{\ln a} + C$

Example 7:

a. $\int 5e^x \, dx = 5e^x + C$

b. $\int 6^x \, dx = \frac{1}{\ln 6} \cdot 6^x + C = \frac{6^x}{\ln 6} + C$

Recall these properties of integrals from sec 6-6:

Properties of Integrals	
$\int kf(x)\,dx = k\int f(x)\,dx$	where k is any constant
Addition rule: $\int [f(x)+g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx$	
Subtraction rule: $\int [f(x)-g(x)]\,dx = \int f(x)\,dx - \int g(x)\,dx$	

Example 8:

a. $\int \left[x^3 - 2x^2 + \frac{4}{x} - 1\right] dx = \frac{1}{4}x^4 - \frac{2}{3}x^3 + 4\ln|x| - x + C$

$4 \cdot \frac{1}{x}$

b. $\int \left[2\cos x - \frac{3}{x} + \frac{5}{\sqrt{x}}\right] dx = 2\sin x - 3\ln|x| + 10x^{1/2} + C$

$-3 \cdot \frac{1}{x}$ $\quad 5x^{-1/2}$

5.2

Strategies for Integration

1. When integrating, we often need to **rewrite** or **simplify** the integrand BEFORE attempting to integrate.
2. Don't **jump** into integration-be on the lookout to see if the problem can be simplified FIRST!!!
3. There are no **product** and **quotient** rules like we do for derivatives, so, these types of problems need to be simplified first!
4. Know your derivative rules! Remember that an antiderivative is the **reverse** of a derivative. Use this to recognize common derivatives within an integration problem.

Example 9:

a. $\int (3x-1)(x+4)dx = \int (3x^2+11x-4)dx$

$= x^3 + \frac{11}{2}x^2 - 4x + C$

b. $\int \cos^2 x \sec x \, dx = \int \cos x \, dx$ $\qquad \cos x \cdot \cancel{\cos x} \cdot \frac{1}{\cancel{\cos x}}$

$= \sin x + C$

c. $\int (2x-3)^2 dx = \int (4x^2 - 12x + 9)\, dx$ $\qquad (2x-3)(2x-3)$

$= \frac{4}{3}x^3 - 6x^2 + 9x + C$

d. $\int \frac{x^2-4x-21}{x+3} dx = \int (x-7)dx$ $\qquad \frac{(x-7)\cancel{(x+3)}}{\cancel{x+3}}$

$= \frac{1}{2}x^2 - 7x + C$

e. $\int \frac{x^2+4x-2}{\sqrt{x}} dx = \int x^{-1/2}(x^2 + 4x^1 - 2)\, dx$

$\sqrt{x} \to x^{1/2}$

$= \int \left(x^{3/2} + 4x^{1/2} - 2x^{-1/2}\right) dx$ $\qquad 2 + -\frac{1}{2}$ $\quad \frac{4}{2} - \frac{1}{2}$

$= \frac{2}{5}x^{5/2} + \frac{8}{3}x^{3/2} - 4x^{1/2} + C$

f. $\int \frac{x^2+3x+7}{4x^2} dx = \int \left(\frac{x^2}{4x^2} + \frac{3x}{4x^2} + \frac{7}{4x^2}\right) dx$

$= \int \left(\frac{1}{4} + \frac{3}{4x} + \frac{7}{4x^2}\right) dx$

$= \frac{1}{4}x + \frac{3}{4}\ln|x| + \frac{-7}{4}x^{-1} + C$ $\qquad \frac{3}{4x} = \frac{3}{4}\left(\frac{1}{x}\right)$ $\quad \frac{7}{4x^2} = \frac{7}{4}x^{-2}$

$= \frac{1}{4}x + \frac{3}{4}\ln|x| - \frac{7}{4x} + C$

The integrals we have been doing in this section are called **indefinite integrals** which find an **antiderivative**. Indefinite integrals are important, but they do not necessarily represent anything tangible. However, if we include limits on the integral, then this becomes a **definite integral**, which represents the net area bounded by the function and the x axis between the limits of integration.

To evaluate **definite integrals**, we will use the Fundamental Theorem of Calculus, Part 1:

Fundamental Theorem of Calculus (part 1)
If the function $f(x)$ is continuous on the closed interval $[a, b]$, then $\int_a^b f(x)dx = F(b) - F(a)$ where $F'(x) = f(x)$ (or in other words, $F(x)$ is an antiderivative of $f(x)$)

How to evaluate Definite Integrals using the FTC...
1. Find the antiderivative (integrate) of the integrand. 2. Apply the FTC by evaluating the antiderivative at the given limits and subtracting. 3. Simplify as needed.

Example 10: $\int_1^5 (4x-3)\,dx = \left[2x^2-3x\right]\Big|_1^5$

$= 50-15-(2-3)$

$= 35-{-1}$

$= 36$

Example 11: $\int_2^4 \left(\frac{1}{x^2} - \frac{3}{x^3}\right)\, dx = \int_2^4 \left(x^{-2} - 3x^{-3}\right) dx$

$$= \left(-1x^{-1} + \frac{3}{2}x^{-2}\right)\Big|_2^4$$

$$= \left(\frac{-1}{x} + \frac{3}{2x^2}\right)\Big|_2^4$$

$$= \frac{-1}{4} + \frac{3}{32} - \left(\frac{-1}{2} + \frac{3}{8}\right)$$

$$= \frac{-1}{32}$$

Example 12: $\int_{1}^{-2}\left(e^{x}+\frac{3}{x}+2\right) = \left[e^{x}+3\ln|x|+2x\right]\Big|_{1}^{-2}$

$= e^{-2}+3\ln 2-4+\left(-e^{1}-3\underbrace{\ln 1}_{\ln 1=0}-2\right)$

$= \frac{1}{e^{2}}+3\ln 2-6-e$

Example 13: $\int_0^{\frac{\pi}{4}} (-2\sin x)\,dx = +2\cos x \Big|_0^{\frac{\pi}{4}}$

$= 2\cos\frac{\pi}{4} - 2\cos 0$

$= 2\cdot\frac{\sqrt{2}}{2} - 2\cdot 1$

$= \sqrt{2} - 2$